

Agricultural Experimental Design – Complete Notes

1. PURPOSE OF EXPERIMENTATION

Agricultural experiments are conducted to generate reliable, scientific evidence for decision-making.

Main Objectives

- **Compare treatments**
 - Example: fertilizers, crop varieties, irrigation methods, pesticides
- **Understand interactions**
- **Optimize production**
- **Support real-world decisions**

2. CORE PRINCIPLES OF DESIGN

2.1 Replication

Repeating each treatment multiple times.

Purpose

- Estimate **experimental error**
- Increase **precision**
- Improve **statistical power**

Key Insight

- More replication → better detection of real differences
- But:
 - Too much replication = waste of resources
 - Too little replication = unreliable results

2.2 Randomization

Assign treatments to experimental units randomly.

Purpose

- Prevent bias (intentional or unintentional)
- Ensure validity of statistical tests

Reality Check

Without randomization:

- Your experiment becomes **biased**
- Your ANOVA becomes **invalid**

2.3 Local Control (Blocking)

Grouping similar experimental units into blocks.

Within Block	Between Blocks
Homogeneous	Heterogeneous

Purpose

- Reduce variability caused by environmental factors
- Improve precision of treatment comparison

Example

Field with fertility gradient → divide into blocks across gradient

👉 Truth:

Blocking is often more powerful than adding replication.

3. STATISTICAL FOUNDATION

3.1 Hypothesis Testing

- **Null Hypothesis (H_0):** All treatments are equal
- **Alternative Hypothesis (H_a):** At least one treatment differs

3.2 F-Test

$$F = \frac{MS_{Treatment}}{MS_{Error}}$$

What this formula REALLY means

You are comparing two types of variation:

(A) MS_t (Treatment Variation)

- Variation between treatment means

- Comes from:
 - fertilizer differences
 - variety differences
 - irrigation differences

👉 This is the signal

(B) MS_e (Error Variation)

- Variation within the same treatment
- Comes from:
 - soil variation
 - measurement error
 - random noise

👉 This is the noise

Core Idea: “Is the signal (treatment effect) stronger than the noise (random error)?”

3.3 Degrees of Freedom (DF)

Definition:

Amount of independent information available

Importance

- Determines accuracy of variance estimates
- Influence’s reliability of statistical tests

Key Insight

- Low DF → unstable estimates → weak conclusions

3.4 Errors in Hypothesis Testing

Type	Meaning
Type I Error (α)	False positive (detect effect when none exists)
Type II Error (β)	False negative (miss real effect)

Trade-off

- Lower $\alpha \rightarrow$ increases β
- Balance is required

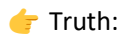
3.5 Statistical Power

Definition:

Probability of detecting a true treatment effect

Influencing Factors

- Replication $\uparrow \rightarrow$ Power $\uparrow$
- Variance $\downarrow \rightarrow$ Power $\uparrow$
- Effect size $\uparrow \rightarrow$ Power $\uparrow$



Truth:

Most agricultural experiments are underpowered.

4. CHOOSING EXPERIMENTAL DESIGN

4.1 Decision Logic

Condition	Recommended Design
Homogeneous area	CRD
One environmental gradient	RCBD
Multiple factors	Factorial
Hard-to-change factor	Split-Plot
Large-scale intersecting factors	Strip Plot
Many treatments	Alpha Lattice

4.2 Design Comparison

Design	Use Case	Precision
CRD	Lab/controlled	High
RCBD	Field	High
Factorial	Interaction study	High

Split-Plot	Practical constraints	Mixed
Strip Plot	Two large factors	Complex

5. EXPERIMENTAL DESIGNS

5.1 Completely Randomized Design (CRD)

- No blocking
- Suitable for **uniform conditions (lab/greenhouse)**

Advantage

- Simple analysis
- Maximum degrees of freedom

Limitation

- Poor performance in field conditions

5.2 Randomized Complete Block Design (RCBD)

- Includes blocking
- Each block contains all treatments

Use

- Field experiments with variability

Advantage

- Controls field heterogeneity

5.3 Factorial Design

- Studies **two or more factors simultaneously**

Key Feature

- Detects **interaction effects**

Important:

If you ignore interaction → your conclusion can be wrong.

5.4 Split-Plot Design

Level	Precision
Main Plot	Low
Sub Plot	High

Use

- When one factor is difficult to change (e.g., irrigation)

Reality

- Many researchers misuse this design → wrong conclusions

5.5 Strip Plot Design

- Two large-scale factors arranged in strips
- Intersection forms experimental units

Use

- Large machinery experiments

6. FIELD REALITIES

6.1 Border Effect

Problem:

Edge plants behave differently

Solution

Plot Type	Purpose
Gross Plot	Total area
Net Plot	Data collection

6.2 Site Selection

Key Considerations

- Conduct **uniformity trial**
- Avoid extreme zones
- Understand field history

Study more on this.

REML vs ML

- **ML (Maximum Likelihood):** biased for variance estimation
- **REML (Restricted ML):** unbiased → preferred in mixed models

Variance Components

- Partition variance into:
 - Genotype
 - Environment
 - Error

BLUP (Best Linear Unbiased Predictor)

- Used in breeding trials
- Predicts genetic value

Estimability & Contrasts

- Planned contrasts → hypothesis-driven
- Unplanned → post hoc

🔗 Without this, your ANOVA interpretation is shallow.